

Identifying Maternal Health Access and Risk Patterns Across U.S. Counties

County-level screening with CDC WONDER Natality and HRSA AHRF data

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The decision problem

Where do adverse birth outcomes and measured access constraints overlap?

Public health agencies need a consistent way to focus limited review capacity. This analysis provides a county screening layer for deeper local investigation.

INTENDED USE

Screening for human review, not automatic allocation or individual prediction

578

named counties in the analytic file

50

states represented

14.4M

births reported, 2020-2024

Data and reproducible method

CDC WONDER

Birth weight, prenatal care, tobacco use, and maternal age pooled across 2020-2024.

HRSA AHRF

County workforce, hospitals, population, insurance, poverty, density, and shortage status.

County join

Five-character county FIPS. Unidentified county groups were excluded from the linked model.

1

Prepare

Pool counts and calculate percentages

2

Engineer

Create rates and log density

3

Explore

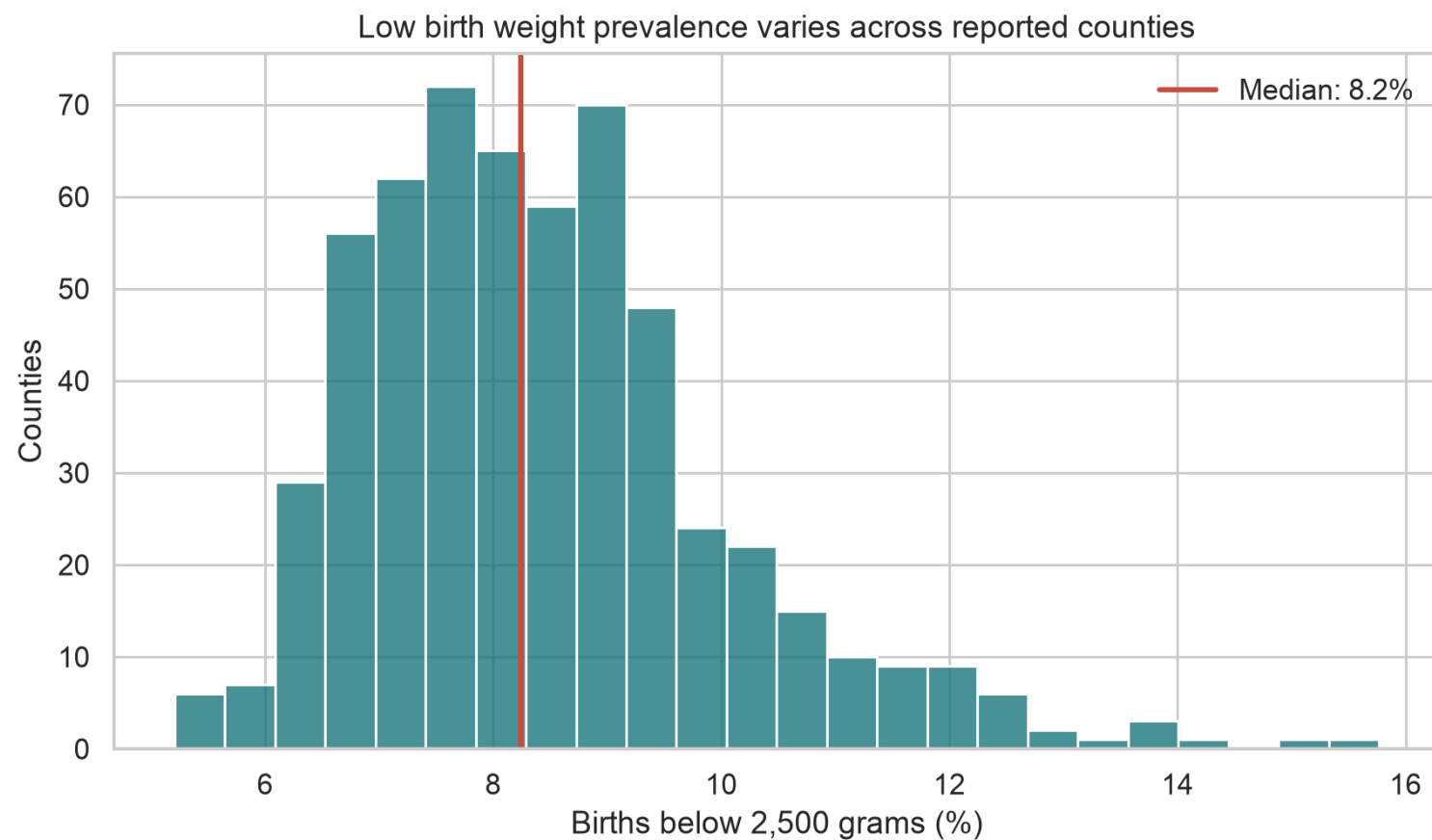
Describe variation and relationships

4

Model

10-fold CV: ridge and random forest

County outcomes vary meaningfully

**8.44%**

county mean

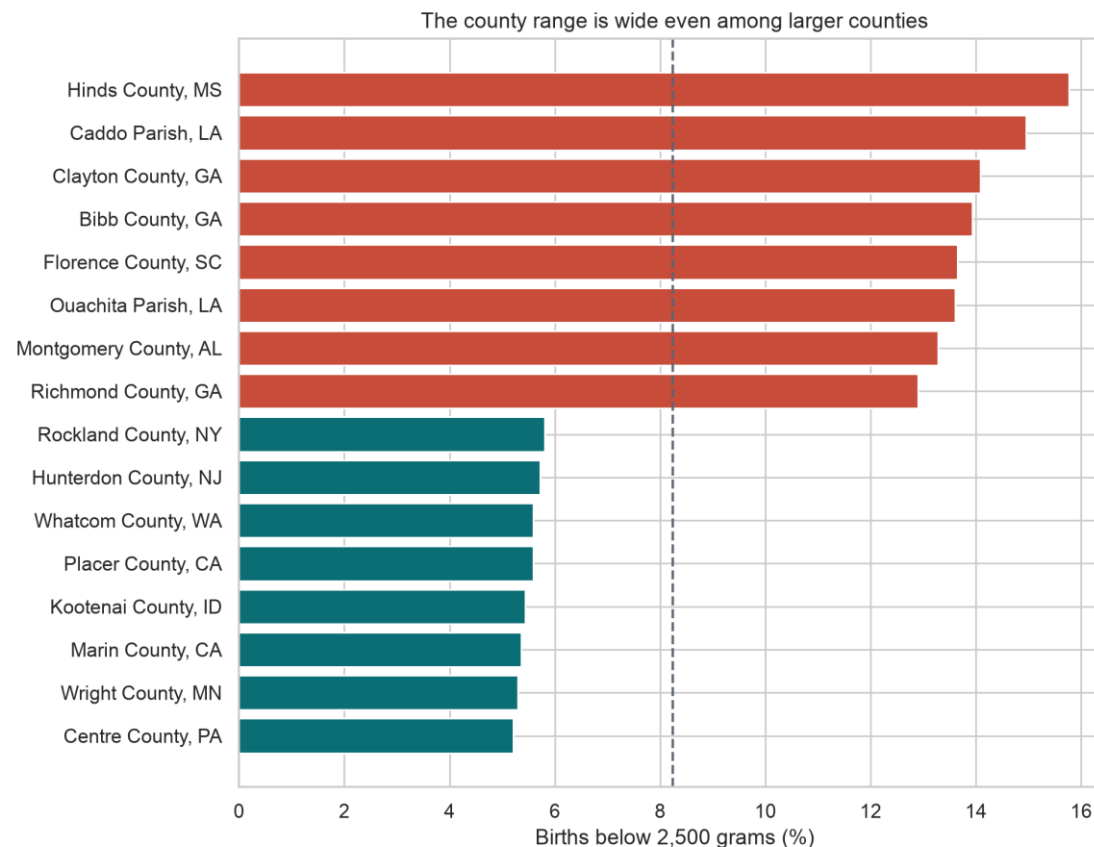
8.24%

county median

5.21-15.77%

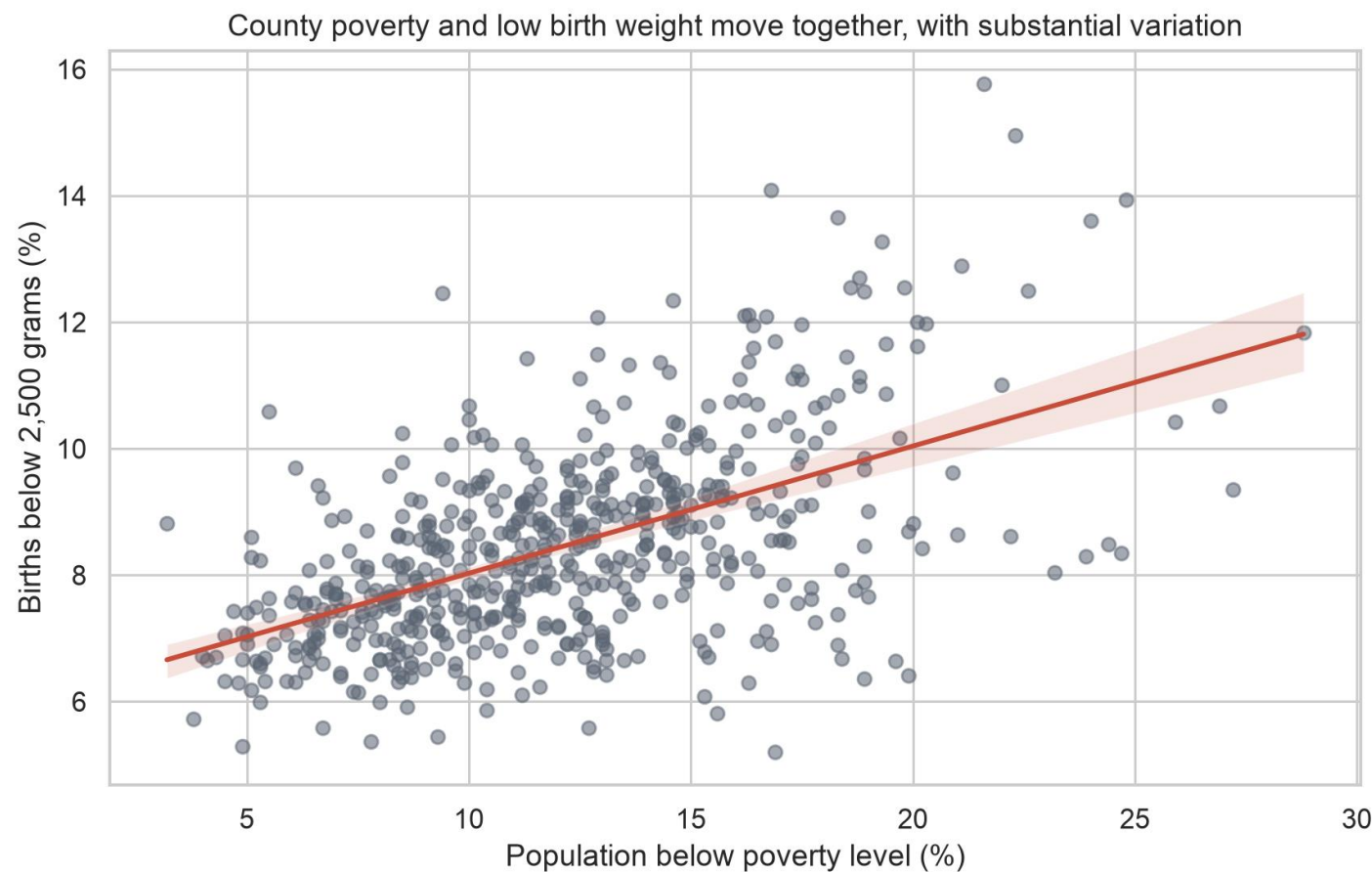
observed county range

Rankings are a starting point



Interpret with count volume, uncertainty, reporting quality, referral patterns, and local context.

Poverty adds signal, not a complete explanation



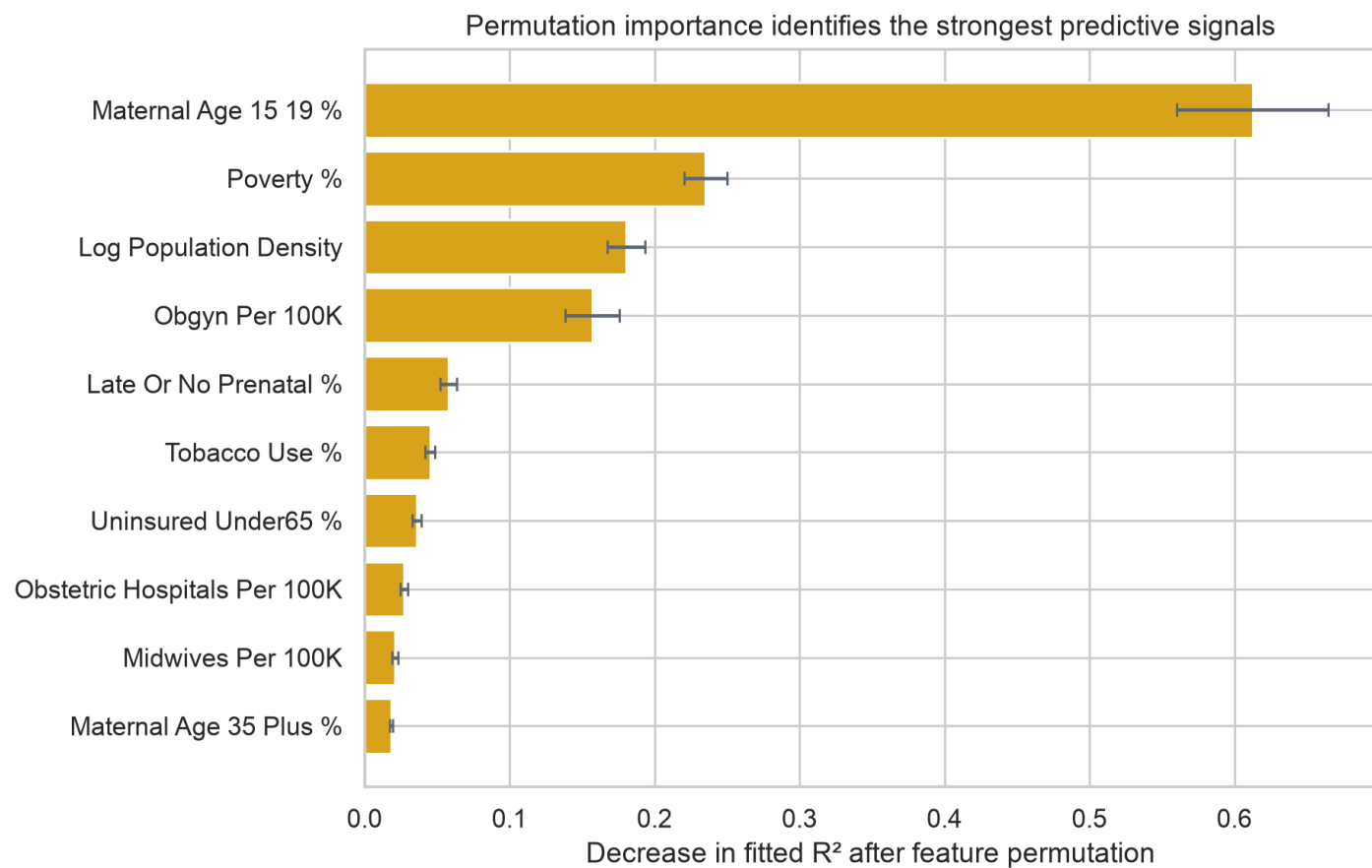
Clear pattern

Higher county poverty is associated with higher low birth weight prevalence in the pooled sample.

Meaningful spread

Counties at similar poverty levels still differ, supporting a multivariable model and local validation.

The model supports screening

**0.571**

10-fold cross-validated R-squared

0.78 pp

mean absolute error

CAUSALITY CHECK

Importance measures predictive contribution. It does not establish cause.

Recommended implementation

1 Refresh

Validate extracts, definitions, exclusions, and dates.

2 Screen

Shortlist counties where outcomes and constraints overlap.

3 Validate

Review local services, travel patterns, and community context.

4 Act and monitor

Select targeted actions and track transparent measures.

RECOMMENDATION

Use model output to direct human review, then choose actions with local partners.

CONCLUSION

Use the evidence with guardrails

A reproducible screening model can focus attention. Local evidence and community review must guide action.

Human review

Confirm data quality, services, and context before action.

Transparent limits

State suppression, exclusions, timing, and uncertainty.

Community voice

Include local partners and a correction process.

Thank you

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